DATABASE CONCEPTS ASSIGNMENTS

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Part – A

Data Understanding

I'm building a database to analyze COVID-19 vaccination data globally and within the US at the state level. My data sources are a collection of CSV files that cover global vaccination records, country data, vaccine details, manufacturers, age-specific data, and additional US state data.

The goal is to organize this information into a database that's easy to query and analyze, providing insights into vaccination trends, manufacturer contributions, and demographic patterns.

2. Identifying Key Entities

After reviewing the data, I identified the main entities:

- **Country**: Represents global vaccination data.
- **US_State**: Specifically tracks vaccination data at the US state level.
- **Manufacturer**: Captures details of companies making the vaccines.
- **Vaccine**: Lists the vaccines used globally.
- **Vaccination Record**: Tracks detailed vaccination numbers by country.
- **Age Group**: Breaks down vaccination data by age.

These entities will form the core of the database.

3. Goals of the Database

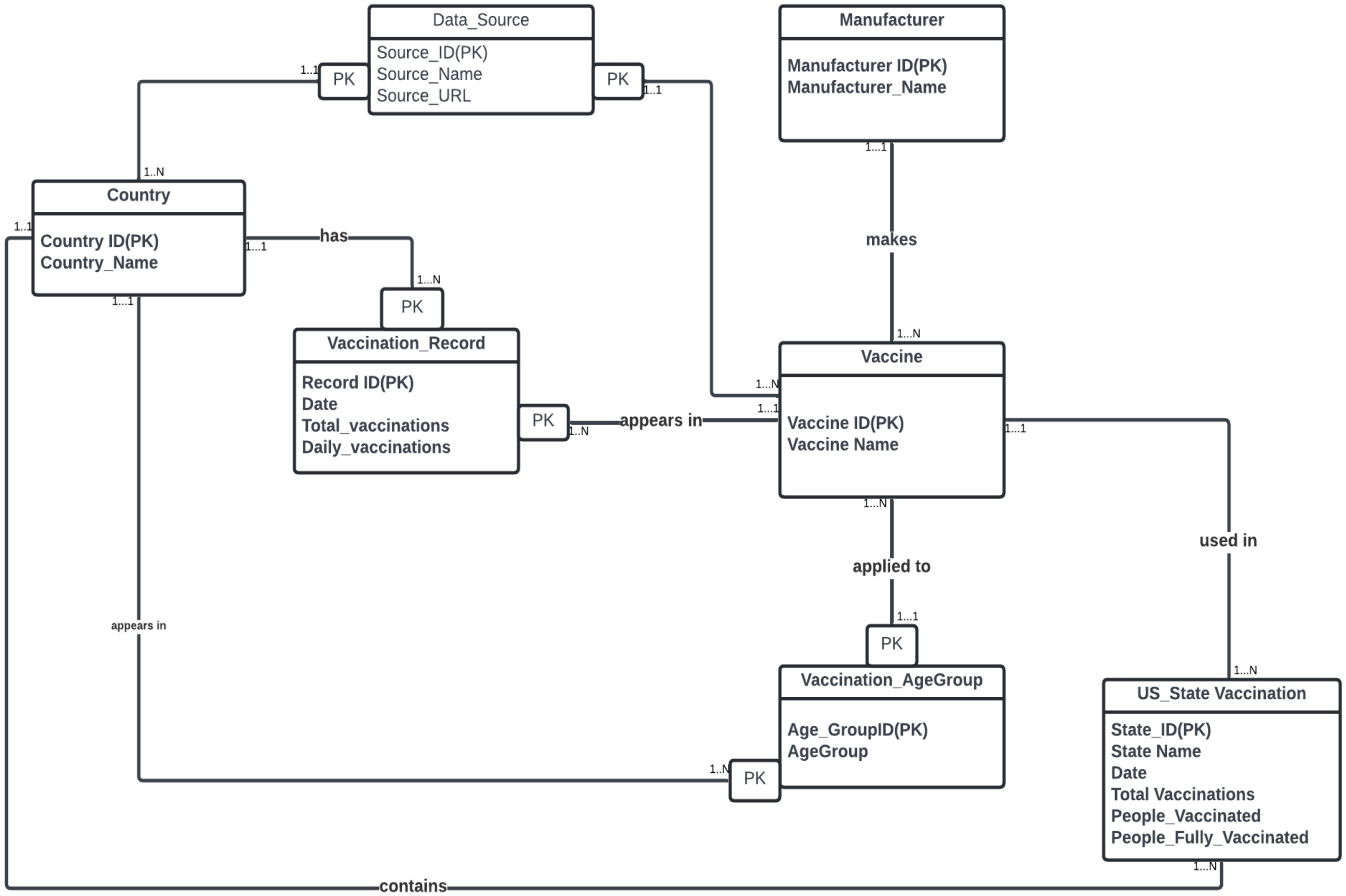
I want this database to:

- Store global and US state vaccination data.
- Support analysis by country, age group, and vaccine type.
- Link vaccines to their manufacturers.
- Provide insights into vaccination trends worldwide, with a deeper dive into US states.

Part-B

ERD Explanation

This ER diagram outlines a global vaccination tracking system. It features strong entities, which stand alone, and weak entities, which rely on others for context.



Here's a quick breakdown:

Strong Entities

1. Country

   Attributes: Country_ID (PK), Country_Name

   Key Relationships:

   - Tracks multiple vaccination records.
   - Includes data for various age groups and state-level entries.
   - Provides multiple data sources for vaccine information.

2. Manufacturer

   Attributes: Manufacturer_ID (PK), Manufacturer_Name

   Key Relationship:

   - Produces multiple vaccines.

3. Vaccine

   Attributes: Vaccine_ID (PK), Vaccine_Name

Key Relationships:

- Applied across various age groups.
- Used in multiple vaccination records.
- Linked to different data sources and state-level vaccination data.

4. US_State_Vaccination

   Attributes: State_ID (PK), State_Name, Date, Total_Vaccinations, People_Vaccinated, People_Fully_Vaccinated

   - Represents vaccination efforts in US states.

Weak Entities

1. Vaccination_Record

   Attributes: Record_ID (PK), Date, Total_Vaccinations, Daily_Vaccinations

   - Relies on both country and vaccine data.

2. Vaccination_AgeGroup

   Attributes: Age_GroupID (PK), AgeGroup, Vaccination_Administered

   - Contains age-specific vaccination data, dependent on country and vaccine.

3. Data_Source

   Attributes: Source_ID (PK), Source_Name, Source_URL

   - Relies on connections to country and vaccine, detailing the origin of the vaccination data.

Conclusion

This ERD effectively illustrates the connections between countries, vaccines, manufacturers, and vaccination data sources. Strong and weak entities are clearly defined, showing dependencies and the flow of information. The design highlights the relationships and keeps data well-organized, reducing redundancy and maintaining clarity.

Assumptions

1. **Identifying Countries:**

   - I've assigned each country a unique Country_ID, assuming that every Country_Name is also unique to avoid duplicates.

2. **Vaccines and Manufacturers:**

   - Each vaccine is identified by a unique Vaccine_ID and has only one manufacturer. In this dataset, no two manufacturers produce the same vaccine.

3. **Recording Vaccinations:**

   - Every vaccination record has its own Record_ID and is linked to a specific country, vaccine, and date. A country can have many records over time, but each entry is unique to one country, one vaccine, and a particular date.

4. **Age Group Data:**
    - Vaccination data by age group is tracked separately for each country and vaccine, with Vaccination_AgeGroup linking this info back to the country and vaccine.
5. **Normalization:**
    - I cleaned up the original dataset to fit 3rd Normal Form (3NF), getting rid of redundancies and organizing the data efficiently. This included adding a Data_Source entity for clarity.
6. **Data Sources:**
    - Each data source has a unique Source_ID and is tied to a specific country and vaccine, making it easy to trace where vaccination data comes from.
7. **Data Integrity:**
    - I've enforced relationships between countries, vaccines, manufacturers, and records to keep everything consistent. This ensures that all vaccination data, including age-specific and source information, is connected to valid entries.

Normalization Process and Challenges

Initial Design

Initially, the dataset from the CSV files was in a flat format, with fields like Country, Location, Date, Vaccine Type, and various vaccination metrics bundled together. This structure posed risks of data inconsistencies and redundancies if used directly in a relational database. The goal was to normalize the data to ensure consistency and efficiency.

Challenges and Solutions

Step 1: First Normal Form (1NF) - Eliminating Grouped Data

- Problem: Some fields had grouped data, with multiple values in a single cell. For example, Location and Vaccine information were sometimes combined, violating the principle of atomicity (each cell should have a single value).
- Solution: I split grouped data into separate fields, ensuring each cell contained a distinct value. Location and Vaccine were separated into different columns to maintain atomicity.
- Outcome: The data met 1NF standards, eliminating grouped data and defining each field clearly.

Step 2: Second Normal Form (2NF) - Eliminate Partial Dependencies

- Problem: Certain attributes, like Total Vaccinations and People Vaccinated, were only partially dependent on parts of a composite key, leading to potential inconsistencies during updates.
- Solution: I defined a composite primary key in the Vaccination_Record table using Location, Date, and Vaccine. Non-key fields were ensured to rely on the full composite key, eliminating partial dependencies.
- Outcome: The database adhered to 2NF, with all non-key attributes fully dependent on the composite key.

Step 3: Third Normal Form (3NF) - Remove Transitive Dependencies

- Problem: Some fields, like source_name and source_url, were not directly related to the primary key, leading to transitive dependencies. These fields were linked to the data source, not the vaccination data.
- Solution: I created a separate Source table to store source_name and source_url. This table was linked to the vaccination data using Location and Vaccine, eliminating indirect dependencies.
- Outcome: The data achieved 3NF, with all non-key attributes directly linked to the primary key and no transitive dependencies.

Final Design

The final database design includes well-defined tables such as Country, Vaccination_Record, Vaccine, Manufacturer, Vaccination_AgeGroup, and US_State_Vaccination. Each table conforms to 3NF standards, minimizing redundancy, preventing anomalies, and making the data easier to manage and query.

Database Schema

The database schema is designed based on the normalized structure and the entities defined in the ER diagram. The following tables outline the structure for each entity, including primary and foreign keys.

- Country(<u>Country_ID</u>, Country_Name)
- Manufacturer(<u>Manufacturer_ID</u>, Manufacturer_Name)
- Vaccine(<u>Vaccine_ID</u>, Vaccine_Name, Manufacturer_ID*)
- Vaccination_Record(<u>Record_ID</u>, Country_ID*, Vaccine_ID*, Date, TotalVaccinations, DailyVaccinations)
- Vaccination_AgeGroup(<u>Age_GroupID</u>, Country_ID*, Vaccine_ID*, AgeGroup)
- US_State_Vaccination (<u>State_ID</u>, Country_ID*, State_Name, Date, Total_Vaccinations, People_Vaccinated, People_Fully_Vaccinated)
- Source(<u>Source_ID</u>,<u>Country_ID</u>*, <u>Vaccine_ID</u>*, Source_Name, Source_URL)

The Source table has a composite key (Country_ID, Vaccine_ID), meaning both fields together uniquely identify a record in the table.